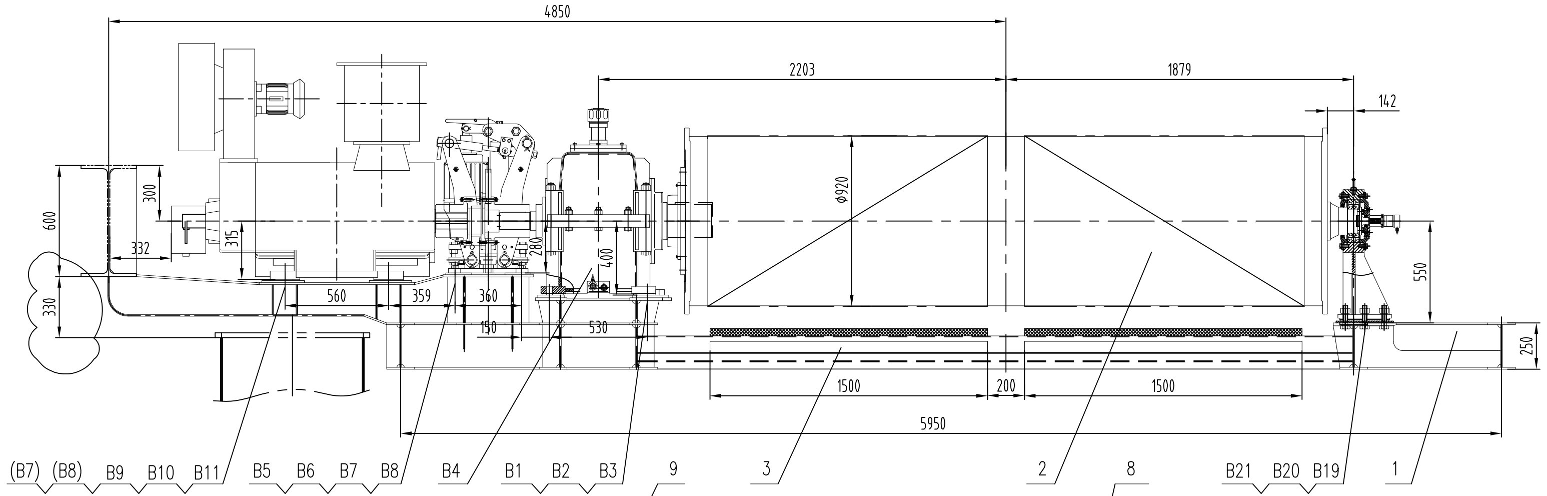
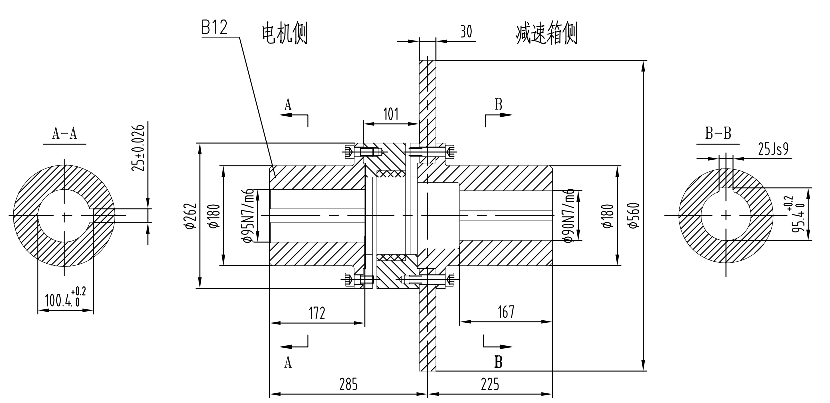
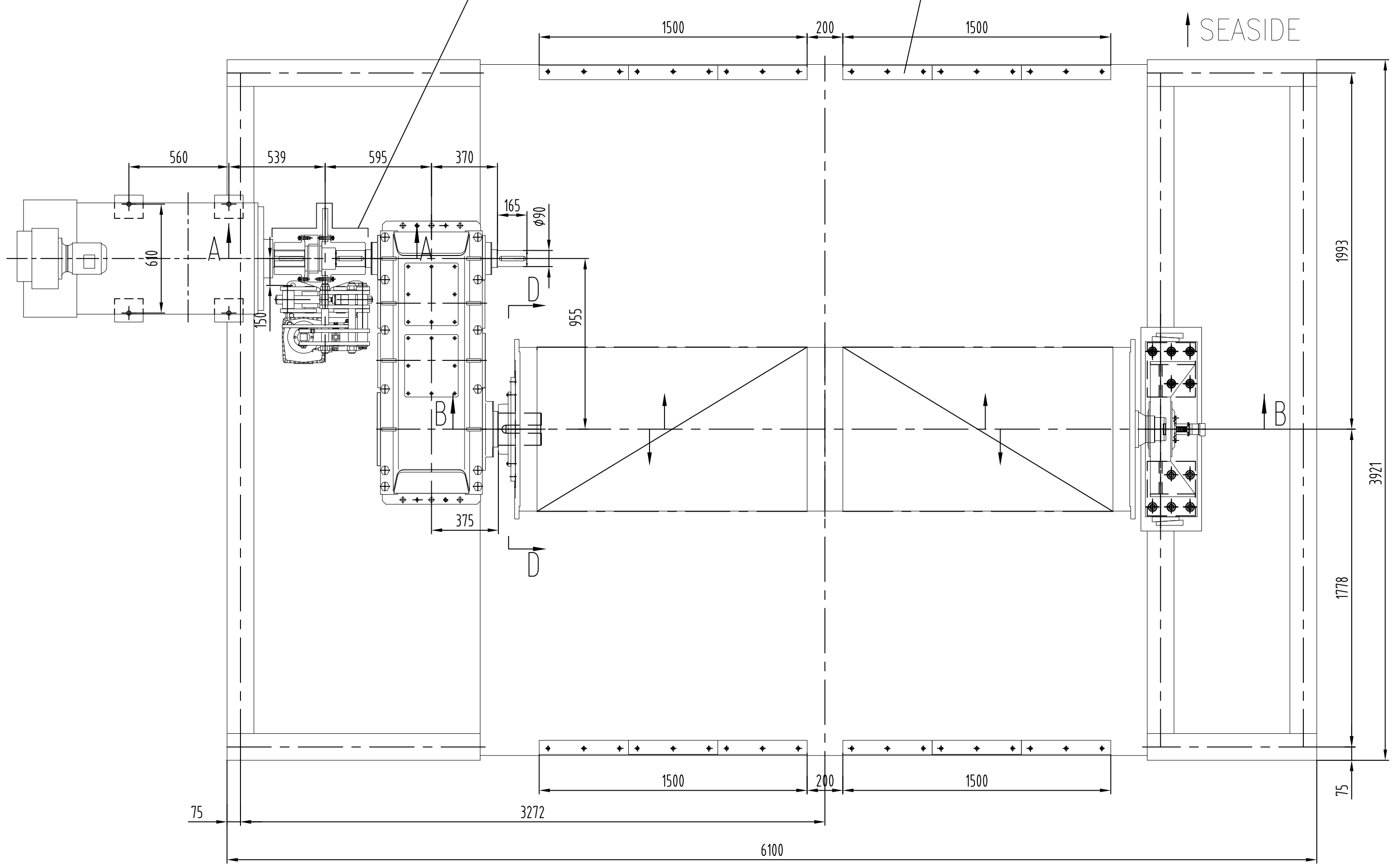
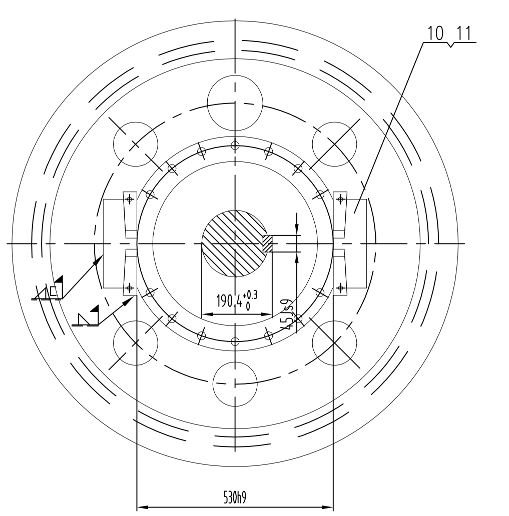


Section A-A





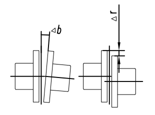
Section D-D



技术要求 Notes

1.电机中心线与减速箱中心线的交角(△b)应小于2′,径向相对偏差(△r)应小于0.05mm,见右图

1. The angular displacement (△b) between the axes of the high speed shaft of reducer and motor shall less than 2′ and the radial misalignment (△r) shall less than 0.05mm. Please refer to figure right.



2.卷筒中心线与减速箱低速轴中心线的交角应小于10′.

2. The angular displacement between the axes of drum and low speed shaft of reducer shall less than 10′.

3.装配时,轴承,齿形联轴器应加适量润滑脂.

3. Proper amount of grease shall be added into the bearings and gears of coupling.

4.卷筒联轴节上的锲块须保证与联轴节和外侧锲形档块均有效接触.

4. Taper blocks on drum coupling shall be tightly contracted with the coupling and outer taper shear blocks.

5.减速器等零部件装配完成后,定位装置应全部有效固定,定位面有效接触,内侧定位块点焊,锁紧螺母拧紧.

5. The alignment elements shall be well secured after machinery alignment: locating face contract each other sufficiently, inside taper blocks be tack welded, lock nuts be tightened.

6. 主要螺栓的预紧力见下表,卷筒制动盘,低速联轴节固定螺栓须在螺杆上均匀涂上二硫化钼润滑脂.

6. Tighten the bolts to the specified tension force as shown in below table.
The bolts of drum coupling or drum brake disc shall be painted MoS2 on the screw.

7.小车在最大前后伸距时卷筒上至少应保留压绳圈2圈,安全圈3圈,海陆侧绳间留2个空圈.

7. Two clamped and three unclamped full dead wraps shall be remaining on the drums when the trolley is at outreach/backreach. Two empty grooves shall be remaining between two wire ropes.

8.联轴节,制动器等外购零部件安装时应遵守相关零部件使用说明书中的安装要求.

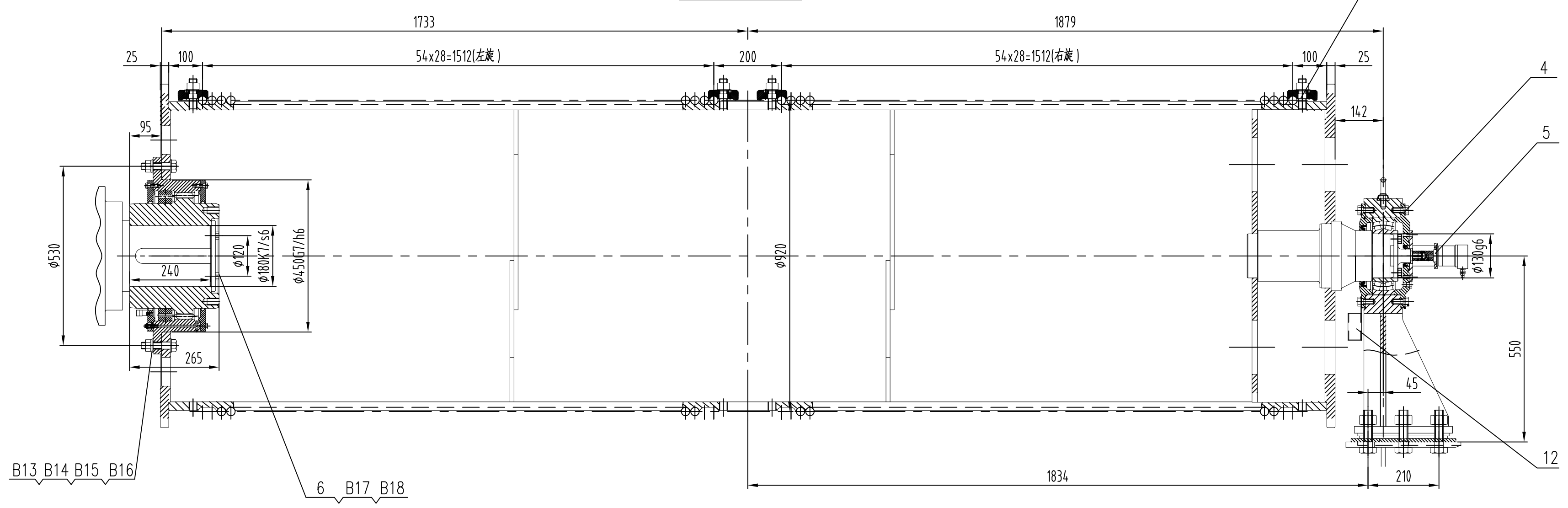
8. Installing instructions in the manuals of purchased parts such as coupling, brake and etc shall also be comformed.

螺栓尺寸D size	M20(10.9s)	M24(8.8s)	M30(8.8s)	M36(8.8s)
预紧力Fo tighten force	156KN	152KN	245KN	358kN
拧紧力矩Mo tightening torque	460Nm	477Nm	954Nm	1665Nm
扭矩系数k torque coefficient	0.15	0.15	0.15	0.15

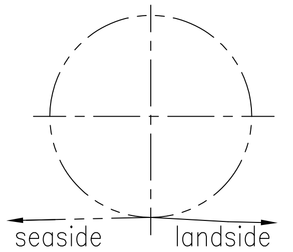
Main Data 主要参数

小车速度 trolley speed		244m/min
电机motor (YASKAWA)	型号 type	BDK-IKMU
	功率 power	270 kw
	转速 speed	1750 rpm
减速器gear box (ZPMC)	型号 type	FH955.21.A6A-00
	速比 ratio	20.702
高速制动器 brake (ZPMC)	型号 type	YP31-1250-560X30
	制动扭矩 brake torque	3550 Nm
	设定力矩 setting torque	2800 Nm
高速联轴节 coupling (ZPMC)	型号 type	MLPK10-560x30
	额定扭矩 torque	5000 Nm
卷筒联轴节 drum coupling (ZPMC)	型号 type	DC05B
	额定扭矩 torque	90 kNm
钢丝绳直径 Dia. of rope		ø25 mm
卷筒名义直径 dia. of drum		ø920 mm

Section B-B



wire rope direction



REV	DATE	BY	DESCRIPTION
01	2012.06.01	YULEI	
02	2012.08.17	YULEI	

描述及日期/Description & Date
修改标志/Sign

设计/DSGN
日期/DATE
校对/CHKD
日期/DATE
审核/EXAM
日期/DATE
图幅/FORMAT A1
材料/MAT Ass. 组件
单重/WEI (Kg) 10059kg
比例/SCALE 1:20
图名/DRAWING NAME Trolley Drive Mechanism 小车牵引机构
图号/DRAWING NO. J526A04
版本/REV 0
页号/PAGE 1

ZPMC
上海振华重工集团

SUBSIDIARY LIST

* The quantity and weight of this list is to one set	PAGE:	1
1 set/crane, 10059kg kg/set, 10059kg kg/crane	TOTAL:	2 PAGE

PROJECT NAME or CODE:	SUB-PART NAME:	SUB DRAW NO:
Sri Lanka CICT 12 STS Cranes	小车牵引机构	J526A04

NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	Q'TY/SET	PIECE WGHT	TOTAL WGHT
1	J526A0401	小车底架	焊接件	1		2896
2	J526A0402	卷筒	焊接件	1		2062
3	J526A0403	集油盘	焊接件	2	30	60
4	J526A0404	卷筒轴承座	焊接件	1		280
5	J526A0405	编码器装置	组件	1		5.9
6	J526A0406	压轴板	Q235	1		3.2
7	RCA06	钢丝绳压板组件6	组件	24	1.4	33.6
8	J526A0407	抗磨块	组件	4	4.8	19.2
9	J526A0408	联轴节罩壳	焊接件	1		11.6
10	ZPDR04-2	楔形挡块	35	2		2.0
11	ZPDR04-1	楔形块	35	4		3.0
12	J526A0409	轴承座集油盘	焊接件	1		1.1

ZPMC	DSGN			TECH				MTRL		
	CHCK			STDD				CHCK		

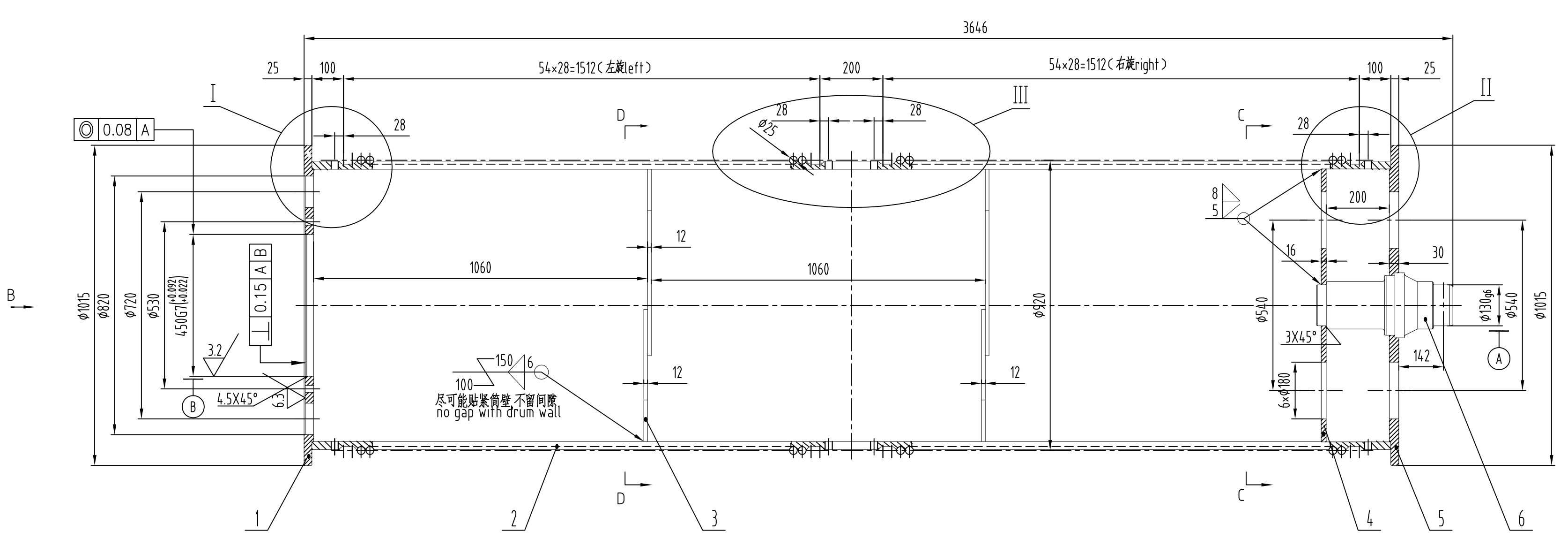
SUBSIDIARY LIST

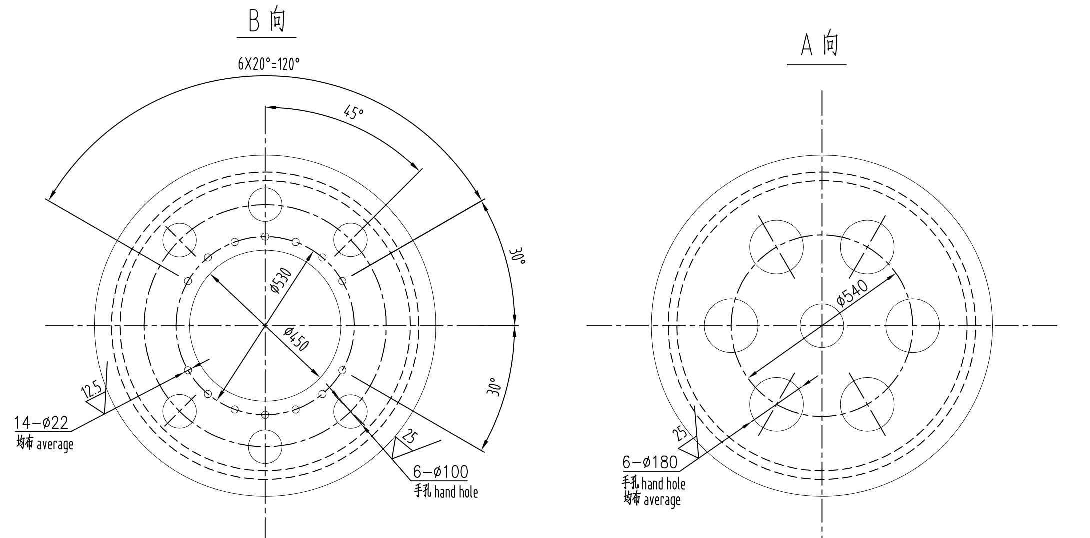
* The quantity and weight of this list is to one set	PAGE:	2
1 set/crane, 10059kg kg/set, 10059kg kg/crane	TOTAL:	2 PAGE

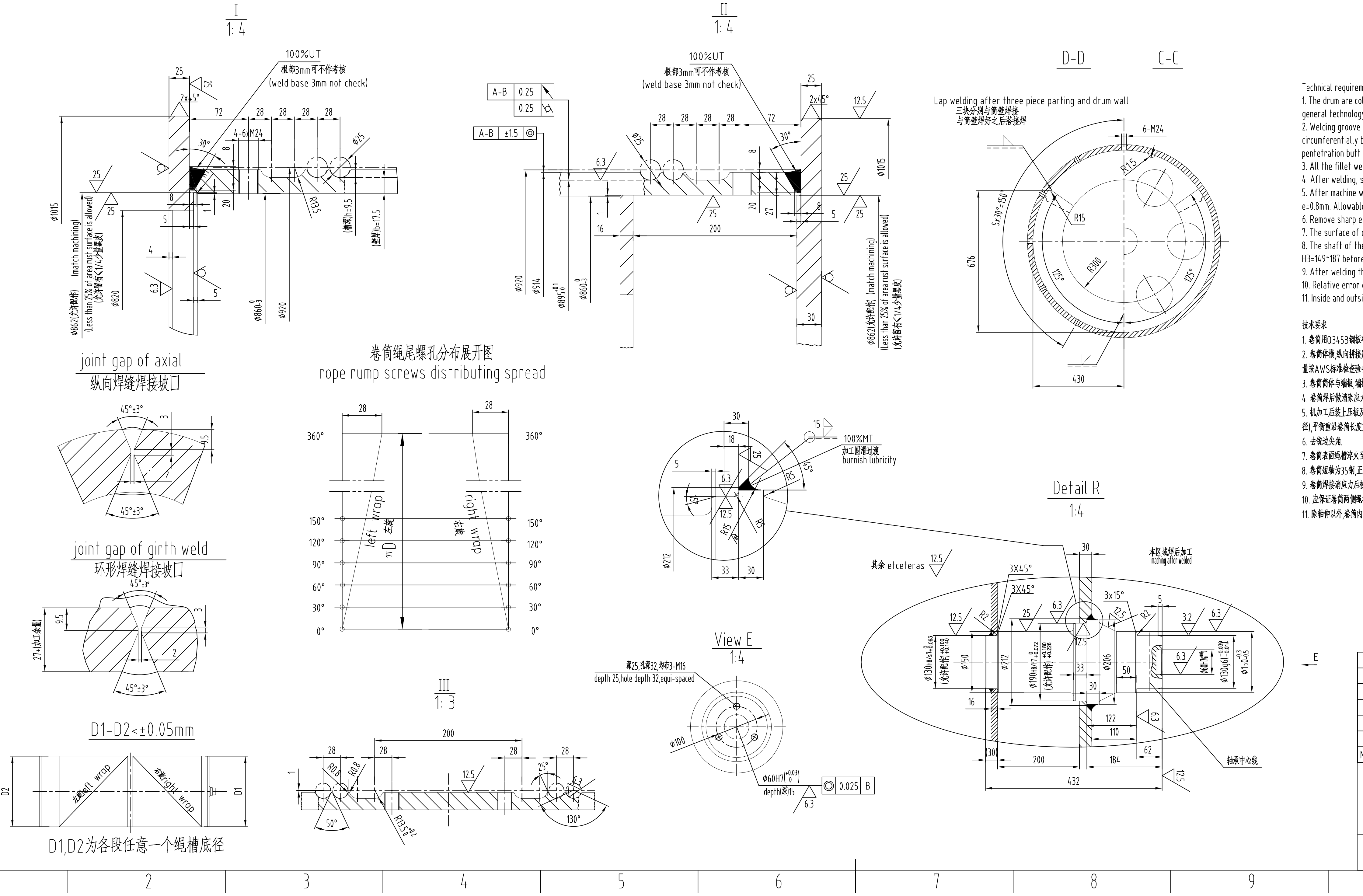
PROJECT NAME or CODE:	SUB-PART NAME:	SUB DRAW NO:
Sri Lanka CICT 12 STS Cranes	小车牵引机构	J526A04

NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	Q'TY/SET	PIECE WGHT	TOTAL WGHT
B1	GB5782-86	螺栓 M30x130	8.8级 达克罗	6	1.2	7.2
B2	ZN6175-T-SL	螺母 M30	8级 达克罗	6	0.35	2.1
B3	GB97.1-85	垫圈 30	200HV 达克罗	6	0.35	2.1
B4	FH955.21.A6A-00	减速箱	ZPMC	1		2370
B5	YP31-1250-560X30	制动器	ZPMC	1		259
B6	GB5782-86	螺栓M24x90	8.8级 达克罗	4	0.4	1.6
B7	ZN6175-T-SL	螺母 M24	8级 达克罗	8	0.11	0.88
B8	GB97.1-85	垫圈 24	200HV 达克罗	8	0.35	2.1
B9	BDK-IKMU	电机	YASKAWA	1		1600
B10	GB5782-86	螺栓 M24x110	8.8级 达克罗	4	0.4	2.4
B11		调整垫片 AS2-24	不锈钢	4	0.4	2.4
B12	MLPK10-560x30	联轴节	ZPMC	1		180
B13	DC05B	低速联轴节	ZPMC	1		245
B14	GB5782-86	螺栓M20x90	10.9级 达克罗	14	0.27	1.62
B15	ZN6175-T-SL	螺母M20	10级 达克罗	14	0.06	1.08
B16	GB97.1-86	垫圈20	300HV 达克罗	28		
B17	GB32.1-88	螺栓M20x35	不锈钢	3	0.06	0.12
B18		不锈钢丝ø1.5	不锈钢	1m		
B19	GB5782-86	螺栓 M27x110	10.9级 达克罗	10	0.27	2.7
B20	ZN6175-T-SL	螺母M27	10级 达克罗	10	0.06	1.08
B21	GB97.1-86	垫圈27	300HV 达克罗	10		

ZPMC	DSGN			TECH				MTRL		
	CHCK			STDD				CHCK		







Technical requirements
1. The drum are cold rolled and welded with steel plate Q345B. Before cold roll,steel plate should be leveled according with general technology.
2. Welding groove preparation shall be made for the longitudinal and transverse butt weld according the drawing. If drum is circumferentially butt welded by subsetions, there shall be >200mm distance between longitudinal welds. All the full pentetration butt welds inspected by 100UT% and one RT per welding according to AWS Welding rods are E5015.
3. All the fillet welding of the drum be inspected by 100%MT. no cracks allowed.
4. After welding, stress release shall be executed.
5. After machine work,drum should be made static balancing test. Offbalance grade is 6.3 grade,allowable eccentricity is e=0.8mm. Allowable mass on drum wall is M=$\frac{1.6G}{Do}$kg(G-drum weight,Do=D-dwr-46),material is Q235.
6. Remove sharp edges.
7. The surface of drum's rope groove will be hardened to HB300~380.
8. The shaft of the drum be made with 35# forged steel,normalizing . It should be inspected by 100%UT,chmical and harder HB=149~187 before welding.
9. After welding the circular degree of drum should <3mm.
10. Relative error of Bottom diamaeter of drum groove<±0.05mm
11. Inside and outside of drum painted by the provision book,except drum shaft.

技术要求
1. 卷筒用Q345B钢板冷卷焊接制成.钢板卷制前必须用通用工艺矫平.
2. 卷筒体横,纵向拼接应按图纸要求开坡口对接.如果采用分段制造环焊对接时,纵向焊缝应错开>200mm.所有对接焊缝做100%UT探伤检验.1.6GDo焊缝质量按AWS标准检查验收,焊条材料为E5015.
3. 卷筒筒体与端板,端板与轴之间的焊缝均进行100%MT探伤检验,不得有任何方向裂纹.
4. 卷筒焊后做消除应力处理.
5. 机加工后装上压板及螺栓作静平衡,不平衡等级G6.3,允许偏心距e=0.8mm.卷筒壁上的允许不平衡质量M=$\frac{1.6G}{Do}$kg(G-卷筒重量,Do-贴平衡重处的直径),平衡重沿卷筒长度方向对称贴,材料为Q235,角焊缝.
6. 去锐边尖角.
7. 卷筒表面绳槽淬火至HB300~380.
8. 卷筒短轴为35钢,正火处理,焊前进行化学成分,硬度(HB=149~187)及超声波探伤检验,达到厂标要求后方可装焊.
9. 卷筒焊接消应力后校圆,圆柱度小于等于3mm.
10. 应保证卷筒两侧绳槽的各段绳槽底径相对误差不大于±0.05mm.
11. 除轴伸以外,卷筒内外部按公司标准涂油漆.

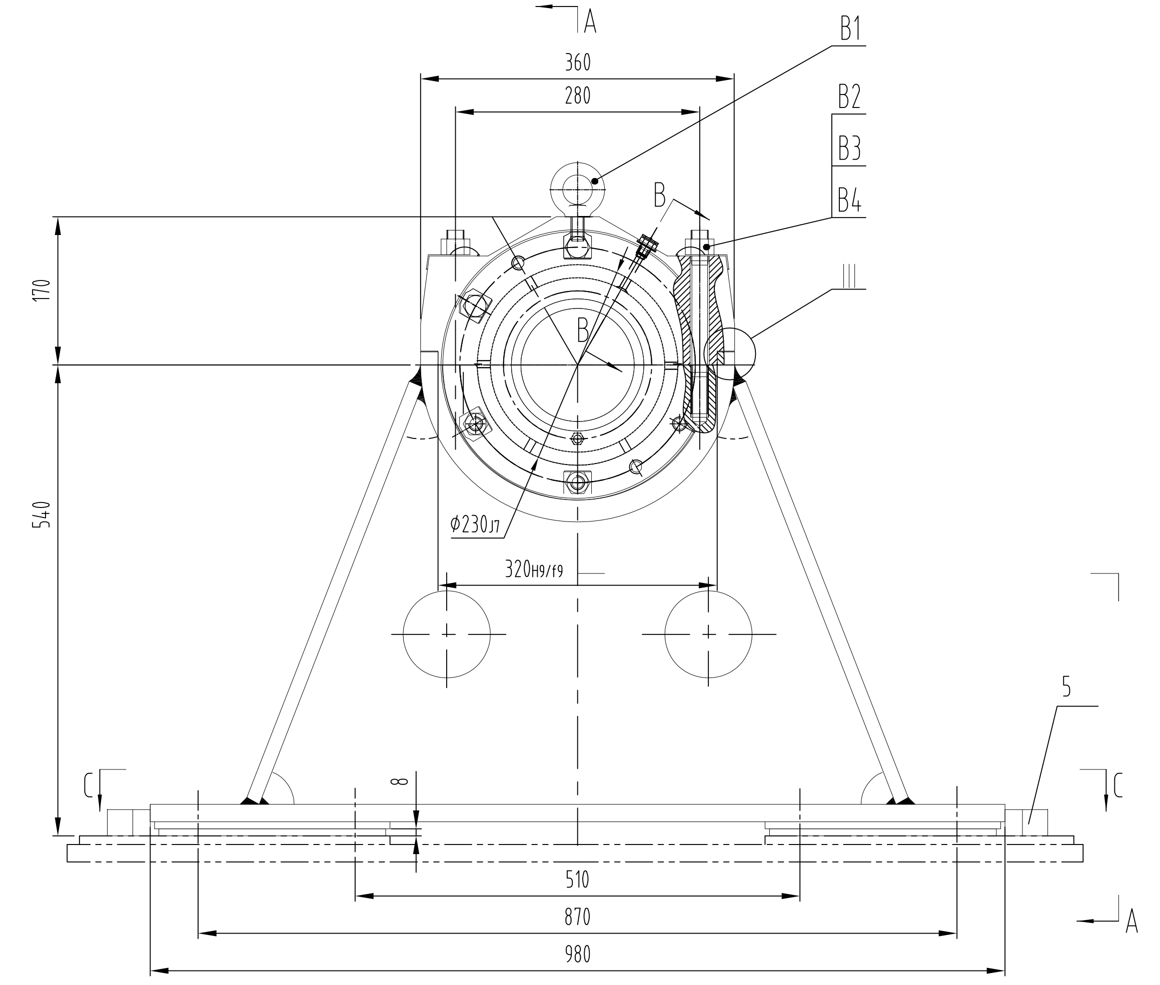
6	GB702-86		圆钢 210 l=432 Round bar	35	1	66.8	66.8	‘Δ ‘
5	GB709-88		钢板 30 F=ϕ1015/ϕ190 Armor plate	Q345B	1	155	155	
4	GB709-88		钢板 16 F=ϕ862/ϕ130 Armor plate	Q345B	1	52	52	
3	GB709-88		钢板 12 F=676×430 Armor plate	Q235	6	16.0	48.0	
2	GB709-88		钢板 (27) F=3424×2810 Armor plate	Q345B	1	1600	1600	
1	GB709-88		钢板 30 F=ϕ1015/ϕ450 Armor plate	Q345B	1	140	140	
NO.	SUB-NO. OR STDD	VER	NAME & SIZE	MTRL	QTY	EACH	TOTAL	NOTE

描述及日期/Description & Date
修改标志/Sign

设计/DSGN	校对/CHKD		图幅/FORMAT A1
日期/DATE	日期/DATE		
材料/MAT 焊接件 Wel.	比例/SCALE 1:10	图名/DRAWING NAME 卷筒 drum	
单重/WEI (Kg) 2062			
ZPMC 上海振华重工集团	图号/DRAWING NO. J526A0402	版本/REV 0	页号/PAGE 1

010330

螺栓规格 Bolt spec.	件号 Item No.	等级 Grade	螺栓预紧力(KN) Pretightening force	螺栓预紧力矩(Nm) Pretightening torque
M20	B4	8.8s	86.2	252(k=0.145)
M27	B8	10.9s	235.2	848(k=0.145)
M16	B12	8.8s	56.5	128(k=0.145)



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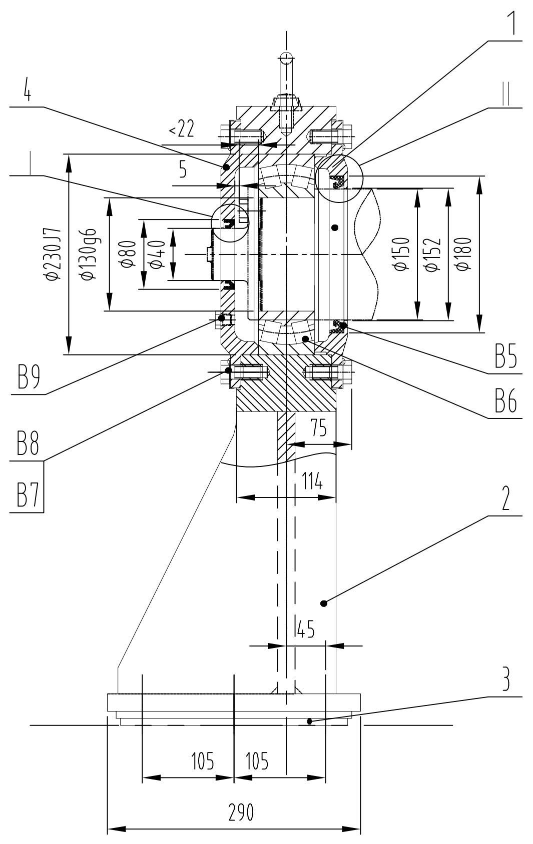
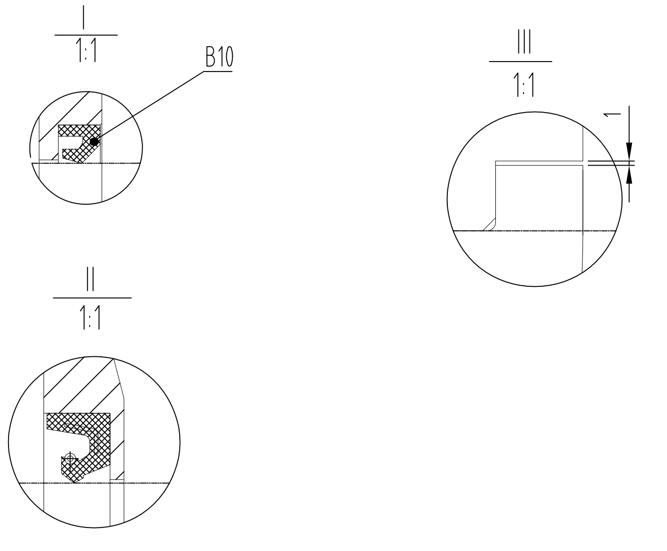
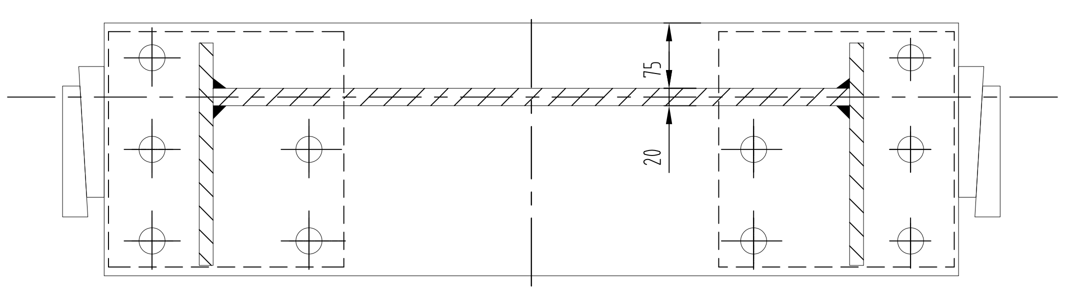
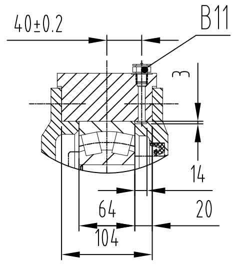




C-C



B-B



Technical Requirements

1. Adjust the height of bearing pedestal to ensure the angular displacement between the axes of drum and low speed shaft of reducer by adjusting plates (item 3). Item 3 could not exceed 2 pieces.
2. Item B9 shall be installed at the bottom of item 4 and can be used as oil outlet.
3. Item B5 and B10 (Oil seals) should be installed according to detail Ⅰ and Ⅱ.
4. Item B1 (screw) shall be used to lift the Bearing Pedestal only.

技术要求

1. 通过垫片(件3)调整轴承座中心高,以保证卷筒中心线与减速箱低速轴中心线交角.调整垫片最多两层.
2. 装配时,件B9(螺塞)布置在件4(端盖II)下部,可作为放油孔.
3. 件B5和B10(骨架密封)的方向必须如图示安装.
4. 件B1(吊环螺钉)仅用于起吊轴承座,不可起吊其它装置.

B11	JISB0203	平头油嘴 Pt1/4"	黄铜	1		0.1	
B10	HG4-692-67	oil seal油封PD40x62x12	成品finished product	1		0.1	
B9	TP1	bolt螺塞 R1/4"	Q235	1		0.1	
B8	GB854-88	Tab-washer止动垫片16	Q215	12	0.01	0.12	Dacro 达克罗
B7	GB5783-86	bolt螺栓 M16x35	8.8级grade	12	0.1	1.2	Dacro 达克罗
B6		bearing轴承 22226CC	Φ130xΦ230x64	1		16	SKF
B5	HG4-692-67	oil seal油封PD150x180x16	成品finished product	1		0.1	
B4	GB900-88	stub 双头螺柱 M20x170	8.8级grade	4	0.5	2.0	Dacro 达克罗
B3	GB854-88	Tab-washer止动垫片20	Q215	4	0.05	0.2	Dacro 达克罗
B2	ZN6175-T-SL	Nut螺目 M20	8级grade	4			Dacro 达克罗
B1	GB825-88	screw吊环螺钉 M16	20	1		0.2	Dacro 达克罗
5	ZP2	locating block2号定位块	Q235	2付	1.1	2.2	
4	J526A040404	end cover端盖2	Q235	1		11.5	
3	J526A040403	adjusting plate调整垫	Q235	2	4.54	9.0	
2	J526A040402	pedestal 轴承座	焊接件welded	1		186	
1	J526A040401	end cover端盖1	Q235	1		9.8	

描述及日期/Description & Date
设计/DSGN
日期/DATE
校对/CHKD
日期/DATE
图幅/FORMAT
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上海振华重工集团
图号/DRAWING NO.
J526A0404
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